

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 1: The Atmosphere: Structure, Composition & ISA

Syllabus Source: CAE Oxford (pp. 1-32) | Pass Target: >= 90% AviationExam

1. COMPOSITION & VERTICAL THERMAL STRUCTURE

The atmosphere is a mechanical mixture of gases surrounding the Earth, structured into concentric thermal layers:

- Gas Composition of Dry Air: Nitrogen (N2) = 78.08%, Oxygen (O2) = 20.95%, Argon (Ar) = 0.93%, Carbon Dioxide (CO2) = 0.04%. Water vapor varies from 0% (polar/desert) to 4% (humid tropics).
- Troposphere: Lowest layer containing 75% of atmospheric mass and virtually all weather (clouds, precipitation). Temperature decreases with altitude. Upper boundary: Tropopause.
- Tropopause Height Variation: Pole: ~8 km (26,000 ft) with temperature ~ -45°C to -50°C. Mid-latitudes: ~11 km (36,090 ft) at -56.5°C. Equator: ~16 to 18 km (55,000 ft) with temperature ~ -75°C to -80°C (COLDEST TROPOPAUSE IS OVER THE EQUATOR!).
- Stratosphere: From Tropopause to Stratopause (~50 km). Temperature increases with altitude due to UV absorption by Ozone (O3). Smooth air, virtually zero water vapor.

2. THE ICAO STANDARD ATMOSPHERE (ISA)

ISA provides an internationally agreed baseline for calibrating altimeters and aircraft performance:

ISA Parameter	Standard Sea Level Value	Lapse Rate / Boundary Specification
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Mean Sea Level Pressure (MSL)	1013.25 hPa (mbar) = 29.92 inHg = 760 mmHg	Decreases ~ 1 hPa per 27 ft (or 30 ft) at sea level.
Mean Sea Level Temperature	+15.0°C (288.15 Kelvin)	Lapses at exactly 2.0°C / 1,000 ft (0.65°C / 100 m).
Mean Sea Level Air Density	1.225 kg/m^3	Decreases with altitude and increasing temperature.
Speed of Sound at MSL (a_0)	340.3 m/s = 661.5 knots	Depends solely on absolute temperature: $a = 38.94 \times \sqrt{T_K}$.
ISA Tropopause Boundary	11,000 meters = 36,090 ft	Temperature reaches -56.5°C (216.65 K), remaining ISOTHERMAL up to 20 km (65,600 ft).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating ISA Temperature & ISA Deviation

SCENARIO (Fundamental AviationExam Question): An airliner is cruising at FL 330 (33,000 ft). The cockpit Outside Air Temperature (OAT) gauge reads -45°C. QUESTION: What is the ISA temperature at FL 330, and what is the ISA Deviation (Delta ISA)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the ISA Standard Sea Level Temperature: - ISA MSL Temperature = +15.0°C. Step 2: Calculate the standard temperature decrease up to FL 330: - ISA Lapse Rate = 2.0°C per 1,000 ft. - Total temperature decrease = $33 \times 2.0^\circ\text{C} = 66.0^\circ\text{C}$. Step 3: Calculate ISA Temperature at 33,000 ft: - ISA Temp = MSL Temp - Temperature Decrease - ISA Temp = $+15.0^\circ\text{C} - 66.0^\circ\text{C} = -51.0^\circ\text{C}$! Step 4: Calculate the ISA Deviation (Delta ISA): - Formula: Delta ISA = Actual OAT - ISA Temperature - Delta ISA = $-45.0^\circ\text{C} - (-51.0^\circ\text{C}) = -45.0^\circ\text{C} + 51.0^\circ\text{C} = +6.0^\circ\text{C}$! FINAL ANSWER: ISA Temperature at FL 330 = -51°C. ISA Deviation = ISA +6°C (WARMER than standard).

[!] EXAM TRAP (AVIATIONEXAM) - Coldest Tropopause Exam Trap

- The tropopause is HIGHEST and COLDEST over the EQUATOR (17 km / -80°C)! - It is LOWEST and WARMEST over the POLES (8 km / -45°C). - Many students incorrectly guess that the poles have the coldest tropopause. That is false!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What are the standard values for temperature and pressure in the ICAO Standard Atmosphere (ISA) at Mean Sea Level?**

[A] 0°C and 1000 hPa [B] +15°C and 1013.25 hPa [C] +20°C and 1013.25 hPa [D] +15°C and 1000 hPa

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The ICAO Standard Atmosphere defines MSL datum as +15°C (288.15 K) and 1013.25 hPa (29.92 inHg).

Q2: Where on Earth is the tropopause physically highest and its temperature coldest?

[A] Over the North and South Poles [B] Over the Equator (height ~16-18 km, temperature ~ -75°C to -80°C) [C] Over mid-latitude depressions [D] Over the Sahara desert

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Intense convective heating expands the equatorial troposphere to 17-18 km. Because temperature lapses continuously through this deep column, the equatorial tropopause is the coldest.

Q3: Up to what altitude does the ISA temperature lapse rate of 2°C per 1,000 ft remain valid before becoming isothermal?

[A] 20,000 ft [B] 36,090 ft (11,000 meters), where temperature reaches -56.5°C [C] 50,000 ft [D] 45,000 ft

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Under ISA definition, the lapse rate terminates at 11 km (36,090 ft) at -56.5°C, where the isothermal stratosphere begins.

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Chapter 2: Atmospheric Pressure, Altimetry & Q-Codes

Syllabus Source: CAE Oxford (pp. 33-68) | Pass Target: >= 90% AviationExam

1. PRESSURE GRADIENTS & ALTIMETER Q-CODES

Atmospheric pressure is the weight of the air column above a unit surface area. It governs altimetry:

Q-Code / Datum	Reference Pressure Datum Plane	Altimeter Reading & Cockpit Usage
Q-Code / Datum	Reference Pressure Datum Plane	Altimeter Reading & Cockpit Usage
QNH (Nautical Height)	Atmospheric pressure reduced to Mean Sea Level (MSL) using ISA lapse rate.	Altimeter reads ALTITUDE above MSL. Reads exact aerodrome elevation on runway!
QFE (Field Elevation)	Actual atmospheric pressure measured at the aerodrome reference datum.	Altimeter reads HEIGHT (AGL) above aerodrome. Reads exactly ZERO on runway threshold!
QNE / Standard (1013.25)	Standard datum plane 1013.25 hPa.	Altimeter reads FLIGHT LEVELS (FL) above Transition Altitude. (e.g. FL 350 = 35,000 ft standard).

2. TEMPERATURE ERRORS IN ALTIMETRY

An altimeter is calibrated strictly to the ISA atmosphere. Non-standard temperatures create dangerous errors:

- Rule of Thumb: 'From High to Low or Hot to Cold, Look Out Below!' When flying into air that is colder than ISA, the atmospheric column compresses. The altimeter over-reads, and your TRUE ALTITUDE IS LOWER than indicated!
- Cold Temperature Correction: True Altitude Correction = 4 ft x (Height above station / 1,000 ft) x (ISA Deviation below standard).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Altimeter Reading Changing QNH to Standard

SCENARIO (Classic Altimetry Transition Drill): An aircraft is cruising at an Indicated Altitude of 5,000 ft with the subscale set to QNH 993 hPa. ATC clears the pilot to climb to FL 80 and instructs: 'Set Standard 1013 hPa'. QUESTION: What will the altimeter indicate immediately after setting the subscale to 1013.25 hPa before any climb takes place? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Pressure Difference on the subscale: - New setting = 1013 hPa - Old setting = 993 hPa - Pressure Change = 1013 - 993 = +20 hPa. Step 2: Understand the physical rule for subscale adjustment: - Turning the subscale to a HIGHER pressure number forces the altimeter hands to WIND UP (indicate a higher altitude)! - Rate of change: 1 hPa = approx 27 ft (or 30 ft in low-altitude rules). - Using standard EASA rule of 27 ft/hPa (or 30 ft/hPa): * 20 hPa x 27 ft/hPa = 540 ft (using 30 ft/hPa = 600 ft). Step 3: Calculate the New Indicated Altitude: - New Indicated Altitude = Old Indicated + Adjustment - Using 27 ft/hPa: 5,000 ft + 540 ft = 5,540 ft! - (Using 30 ft/hPa: 5,000 ft + 600 ft = 5,600 ft). FINAL ANSWER: The altimeter pointer winds clockwise to indicate approximately 5,540 ft to 5,600 ft.

[!] EXAM TRAP (AVIATIONEXAM) - The High to Low Pressure Trap

- If you fly from an area of HIGH pressure into an area of LOW pressure without resetting the altimeter: - The altimeter CONTINUES TO READ THE OLD ALTITUDE, but your airplane has descended! - You are physically CLOSER TO THE GROUND than your instrument displays ('High to low, look out below!').

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: If an aircraft flies from a region of HIGH pressure into an area of LOW pressure without adjusting the altimeter, what does the altimeter read?**

[A] It reads lower than the true altitude [B] It reads HIGHER than the true altitude (the aircraft is lower than the pilot believes) [C] It reads the correct altitude [D] It fails

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The altimeter measures ambient pressure. In low pressure, the altimeter assumes the plane has climbed, so it over-reads. The aircraft's true altitude is lower.

Q2: When on the ground at an aerodrome with the altimeter subscale set to QFE, what does the altimeter indicate?

[A] Aerodrome elevation above sea level [B] Exactly ZERO feet [C] Density altitude [D] Pressure altitude

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. QFE is the measured pressure at the airfield datum. With QFE set on the ground, the altimeter indicates height above the field, which is 0 ft.

Q3: What is the Transition Level (TRL)?

[A] The highest altitude where QNH is used [B] The LOWEST AVAILABLE FLIGHT LEVEL above the Transition Altitude (TA) [C] 3,000 ft [D] The level of the tropopause

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Transition Altitude (TA) is the highest altitude using QNH. Transition Level (TRL) is the lowest usable Flight Level using standard 1013.25 hPa above the transition layer.

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Chapter 3: Atmospheric Density, Temperature & Inversions

Syllabus Source: CAE Oxford (pp. 69-98) | Pass Target: $\geq 90\%$ AviationExam

1. AIR DENSITY & AIRCRAFT PERFORMANCE

Air density (ρ) is mass per unit volume. Lift, drag, engine thrust, and propeller efficiency are directly proportional to air density:

- Density Relationships: Density INCREASES with: Increasing pressure, Decreasing temperature, and Decreasing humidity (dry air is denser than moist air because H₂O molecular weight 18 is lighter than N₂ 28 / O₂ 32!).
- Density Altitude (DA): Pressure altitude corrected for non-standard temperature: Density Alt = Pressure Alt + 120 ft x (OAT - ISA Temp). High density altitude severely degrades take-off roll and climb!

2. TEMPERATURE INVERSIONS (NEGATIVE LAPSE RATE)

An inversion occurs when temperature INCREASES with altitude instead of decreasing:

- Radiation Inversion (Ground Inversion): Forms on clear, calm nights as the ground radiates heat into space. Chills the contact air layer. Disperses mid-morning with solar heating.
- Subsidence Inversion: Forms in high-pressure anticyclones as sinking air warms adiabatically, capping pollution, haze, and stratocumulus clouds below it.
- Frontal Inversion: Occurs across a warm front where warm air ascends over a wedge of cold surface air.
- Aviation Hazards of Inversions: Creates intense low-level windshear across the inversion boundary, severely degrades engine performance climbing through the warm layer, and traps pollutants reducing visibility.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Density Altitude from QNH and OAT

SCENARIO (Classic Performance & Meteorology Exam Question): An airfield has an elevation of 3,000 ft MSL: - Current QNH = 1003 hPa - Outside Air Temperature (OAT) = +31°C QUESTION: What is the Pressure Altitude and the Density Altitude at the airfield? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Pressure Altitude (PA): - Formula: $PA = \text{Elevation} + 27 \text{ ft} \times (1013.25 - QNH)$ - Pressure difference = $1013.25 - 1003 = +10.25 \text{ hPa}$. - Altitude correction = $10.25 \text{ hPa} \times 27 \text{ ft/hPa} = 277 \text{ ft}$ (or ~300 ft). - $PA = 3,000 \text{ ft} + 277 \text{ ft} = 3,277 \text{ ft}$ (~3,280 ft). Step 2: Calculate ISA Temperature at 3,277 ft: - ISA Temp at Sea Level = +15.0°C. - Lapse = $3.28 \times 2^\circ\text{C} = 6.56^\circ\text{C}$. - ISA Temp at airfield = $+15.0^\circ\text{C} - 6.56^\circ\text{C} = +8.44^\circ\text{C}$ (~+8.5°C). Step 3: Calculate ISA Deviation: - Actual OAT = +31.0°C - Delta ISA = $31.0^\circ\text{C} - 8.44^\circ\text{C} = +22.56^\circ\text{C}$ (22.6°C WARMER than standard!). Step 4: Calculate Density Altitude (DA): - Formula: $DA = \text{Pressure Altitude} + 120 \text{ ft} \times \text{Delta ISA}$ - Temperature adjustment = $120 \text{ ft} \times 22.56^\circ\text{C} = 2,707 \text{ ft}$! - $DA = 3,277 \text{ ft} + 2,707 \text{ ft} = 5,984 \text{ ft}$ (~6,000 ft)! FINAL ANSWER: Pressure Altitude = 3,280 ft. Density Altitude = 5,984 ft! (The aircraft will perform as if it were taking off at 6,000 ft altitude!).

[!] EXAM TRAP (AVIATIONEXAM) - Humid Air vs Dry Air Density Trap

- Many students think humid air is heavier than dry air. THIS IS A TRAP! - Water vapor (H₂O, molecular mass 18) is LIGHTER than dry air (N₂ 28 + O₂ 32, average mass 29)! - Therefore, HIGH HUMIDITY DECREASES AIR DENSITY and degrades aircraft performance!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under which weather conditions is a SURFACE RADIATION INVERSION most likely to develop?**

[A] Strong winds and heavy overcast clouds [B] CLEAR NIGHTS with calm or very light wind [C] Active cold front passage [D] Midday summer sun

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Rapid terrestrial radiative cooling into a clear sky combined with calm air allows the surface air to chill rapidly while the air above remains warmer.

Q2: How does an increase in moisture content (relative humidity) affect the density of the air?

[A] Increases density [B] DECREASES density (moist air is less dense than dry air) [C] Has zero effect [D] Doubles air pressure

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Water molecules (H₂O, mol wt 18) displace heavier nitrogen (28) and oxygen (32) molecules in a given volume, making humid air lighter/less dense.

Q3: What flight hazard is most frequently encountered when climbing or descending through a strong temperature inversion?

[A] Severe airframe icing [B] Marked LOW-LEVEL WINDSHEAR and turbulence across the inversion boundary [C] Complete loss of pitot pressure [D] Heavy hail

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The thermal boundary decouples surface air from the upper flow, frequently generating severe vertical and directional windshear.

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Chapter 4: Moisture, Humidity & Atmospheric Stability

Syllabus Source: CAE Oxford (pp. 99-136) | Pass Target: >= 90% AviationExam

1. HUMIDITY CONCEPTS & THE DEW POINT

Water vapor is the driving fuel of atmospheric thermodynamics:

- Relative Humidity (RH): Ratio of actual water vapor pressure to the saturation vapor pressure at that temperature, expressed as %: $RH = (Actual\ Vapor\ Pressure / Saturation\ Vapor\ Pressure) \times 100$.
- Dew Point (DP / Td): The temperature to which unsaturated air must be cooled at constant pressure to become fully saturated (RH = 100%). Small temperature-dew point spread indicates imminent fog or cloud formation!

2. ADIABATIC LAPSE RATES & STABILITY

As an air parcel rises, it expands due to decreasing external pressure and cools without exchanging heat with surrounding air (adiabatic process):

- Dry Adiabatic Lapse Rate (DALR): Applies to unsaturated air: Constant at 3.0°C / 1,000 ft (1.0°C / 100 m = 9.8°C / km).
- Saturated Adiabatic Lapse Rate (SALR): Applies once condensation occurs (saturated air): Variable, approximately 1.5°C / 1,000 ft (0.5°C to 0.7°C / 100 m). SALR is lower than DALR because latent heat of condensation (600 cal/g) is released into the parcel, retarding its cooling!
- Environmental Lapse Rate (ELR): The actual measured temperature profile of the ambient atmosphere.

Atmospheric Stability State	Mathematical Criterion (ELR vs DALR / SALR)	Associated Weather & Cloud Types
Atmospheric Stability State	Mathematical Criterion (ELR vs DALR / SALR)	Associated Weather & Cloud Types
Absolute Stability	ELR < SALR (Lapse rate is shallow or inverted, e.g. < 1.5°C/1,000 ft).	No vertical convection. Stratiform flat clouds (Stratus), smooth air, poor surface visibility/fog.
Absolute Instability	ELR > DALR (Lapse rate is steep, > 3.0°C/1,000 ft / superadiabatic).	Violent convective up/downdrafts, Cumulonimbus (CB), severe turbulence, good visibility outside showers.
Conditional Instability	SALR < ELR < DALR (ELR is between 1.5°C and 3.0°C/1,000 ft).	Stable if air is dry; violently UNSTABLE if lifted to its saturation level (LCL)!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Convective Cloud Base Height

SCENARIO (Extremely Common EASA Exam Question): Surface weather observations

report: - Surface Temperature = +22°C - Surface Dew Point = +12°C - Airfield elevation

= 500 ft MSL QUESTION: At what altitude MSL will the base of convective Cumulus clouds

form? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Temperature - Dew

Point Spread at surface: - Spread = Surface Temperature - Dew Point - Spread =

22°C - 12°C = 10°C. Step 2: Understand how the Spread closes with altitude in a

rising parcel: - Rising dry parcel cools at DALR = 3.0°C per 1,000 ft. - Dew point

also drops slightly as pressure falls = 0.5°C per 1,000 ft. - Net closing rate of

the spread = 3.0°C - 0.5°C = 2.5°C per 1,000 ft (or 400 ft per °C)! Step 3: Calculate

Cloud Base Height Above Ground Level (AGL): - Cloud Base AGL = (Spread / 2.5°C) x

1,000 ft - Cloud Base AGL = (10°C / 2.5°C) x 1,000 ft = 4 x 1,000 ft = 4,000 ft

AGL! Step 4: Convert Height AGL to Altitude MSL (Add airfield elevation): - Cloud

Base MSL = Height AGL + Field Elevation - Cloud Base MSL = 4,000 ft + 500 ft = 4,500

ft MSL! FINAL ANSWER: Cloud base forms at 4,000 ft AGL (4,500 ft MSL).

[!] EXAM TRAP (AVIATIONEXAM) - AGL vs MSL Cloud Base Trap

- The formula $(T - DP)/2.5$ gives cloud base ABOVE GROUND LEVEL (AGL)! - If the exam question asks for ALTITUDE MSL, you MUST ADD aerodrome elevation! - Forgetting to add elevation is the most frequent trap in this question.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Why is the Saturated Adiabatic Lapse Rate (SALR) lower than the Dry Adiabatic Lapse Rate (DALR)?**

[A] Saturated air is heavier [B] Because LATENT HEAT of condensation is released when water vapor condenses, warming the rising parcel [C] Friction with falling rain [D] Sun heating

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. As water vapor condenses into water droplets, latent heat (approx 2.5×10^6 J/kg) is released into the air parcel, partially offsetting adiabatic expansion cooling.

Q2: If the ambient Environmental Lapse Rate (ELR) is 2.2°C per 1,000 ft, what is the stability state of the atmosphere?

[A] Absolute stability [B] Absolute instability [C] CONDITIONAL INSTABILITY (because ELR lies between SALR ~1.5°C and DALR 3.0°C) [D] Neutral

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. When ELR is between SALR and DALR, the atmosphere is conditionally unstable: stable if unsaturated, but unstable if forced to saturation.

Q3: What type of cloud is produced when an absolutely STABLE layer of air is forced to rise over a mountain range?

[A] Cumulonimbus with severe hail [B] STRATIFORM clouds (smooth sheets such as Stratus or Altostratus) without vertical development [C] Isolated towering cumulus [D] Mammatus

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In stable air, displaced air resists vertical motion, producing flat, horizontal layer clouds without thermal turbulence.

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Chapter 5: Wind Dynamics, Coriolis & Local Wind Systems

Syllabus Source: CAE Oxford (pp. 137-174) | Pass Target: $\geq 90\%$ AviationExam

1. FORCES GOVERNING WIND & BALANCE

Air motion is governed by four fundamental forces:

- Pressure Gradient Force (PGF): Acts perpendicular to isobars from HIGH pressure towards LOW pressure. Initiates all horizontal air motion.
- Coriolis Force: Apparent force caused by Earth's rotation. Deflects moving air to the RIGHT in the Northern Hemisphere (to the LEFT in the Southern Hemisphere). Zero at equator; maximum at poles. Proportional to wind speed.
- Geostrophic Wind: Theoretical wind resulting from an exact balance between PGF and Coriolis force blowing parallel to straight, parallel isobars above friction layer.
- Surface Friction Effect (Veering vs Backing): Friction slows surface wind, reducing Coriolis force. PGF pulls air across isobars towards low pressure ($\sim 30^\circ$ over land, $\sim 10^\circ$ over sea). In Northern Hemisphere: Climbing out, wind VEERS (turns clockwise) and INCREASES; Descending to land, wind BACKS (turns counter-clockwise) and DECREASES!

2. LOCAL WIND PHENOMENA

- Sea Breeze vs Land Breeze: Sea breeze develops by day (cooler sea air blows onshore toward heated land). Land breeze develops at night (cooling land air blows offshore).
- Anabatic vs Katabatic Winds: Anabatic wind: Daytime upslope wind as sun heats valley walls. Katabatic wind: Nighttime downslope gravity wind as dense cold air drains into valleys (e.g. Bora, Mistral).
- The Föhn Effect (Chinook): Moist air rises on windward mountain slope cooling at SALR (losing moisture by rain). Descending on leeward slope, dry air warms at DALR ($3^\circ\text{C}/1,000\text{ ft}$), arriving warm and extremely dry!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Temperature & Dew Point in a Föhn Wind

SCENARIO (Standard EASA Föhn Exam Problem): Air at sea level (0 ft) on the windward side of a 6,000 ft mountain has: - Surface Temperature = +18°C - Surface Dew Point = +10.5°C - Cloud base (condensation level) on windward slope = 3,000 ft - Air crosses the ridge at 6,000 ft and descends to sea level on the leeward side QUESTION: What will be the temperature and dew point of the air when it reaches sea level on the leeward side? SOLUTION PASO A PASO (FOR DUMMIES): Part 1: Windward Ascent (0 to 6,000 ft): - From 0 to 3,000 ft (Dry ascent below cloud base at DALR = 3°C/1,000 ft): * Temp decrease = $3 \times 3.0^\circ\text{C} = 9.0^\circ\text{C}$. * Temp at 3,000 ft = $18.0^\circ\text{C} - 9.0^\circ\text{C} = +9.0^\circ\text{C}$. - From 3,000 to 6,000 ft (Saturated ascent inside cloud at SALR = 1.5°C/1,000 ft): * Temp decrease = $3 \times 1.5^\circ\text{C} = 4.5^\circ\text{C}$. * Temp at mountain top (6,000 ft) = $9.0^\circ\text{C} - 4.5^\circ\text{C} = +4.5^\circ\text{C}$. * (Because it is saturated at the peak, Dew Point at 6,000 ft = +4.5°C!). Part 2: Leeward Descent (6,000 ft down to 0 ft): - The air dropped its moisture as rain on the windward side! All clouds evaporate instantly upon descending. - ENTIRE DESCENT IS DRY at DALR = 3.0°C / 1,000 ft! - Temperature increase = $6 \times 3.0^\circ\text{C} = +18.0^\circ\text{C}$. - Final Leeward Temp at sea level = $+4.5^\circ\text{C} + 18.0^\circ\text{C} = +22.5^\circ\text{C}$! Part 3: Final Dew Point on leeward side: - Dew point increases down the slope at 0.5°C / 1,000 ft = $6 \times 0.5^\circ\text{C} = 3.0^\circ\text{C}$. - Final Leeward Dew Point = $4.5^\circ\text{C} + 3.0^\circ\text{C} = +7.5^\circ\text{C}$. FINAL ANSWER: Leeward surface air is dramatically WARMER (+22.5°C vs +18°C) and DRIER (Dew Point +7.5°C vs +10.5°C)! Classic Föhn wind.

[!] EXAM TRAP (AVIATIONEXAM) - Veering vs Backing Memory Trap

- In the Northern Hemisphere: - VEER = Clockwise turn (e.g. 240° to 270°). Occurs when CLIMBING. - BACK = Counter-clockwise turn (e.g. 270° to 240°). Occurs when DESCENDING towards landing.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: In the Northern Hemisphere, what happens to the wind direction and speed when descending from 3,000 ft**

[A] Veers and increases [B] BACKS (turns counter-clockwise) and DECREASES in speed due to surface friction [C] Veers and decreases [D] Remains unchanged

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Surface friction slows the wind, reducing the Coriolis force. The pressure gradient force pulls the wind towards lower pressure, causing it to back and reduce speed.

Q2: What is the primary characteristic of a KATABATIC wind?

[A] Warm daytime wind blowing upslope [B] COLD, DENSE AIR flowing downhill at night under the force of gravity [C] Wind blowing from sea to land [D] Jet stream core

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Nocturnal terrestrial radiation cools mountain air, making it dense and heavy. Gravity pulls this cold pool down valleys and slopes as a katabatic wind.

Q3: In the Föhn effect, why is the air arriving on the leeward side of the mountain warmer than when it started

[A] Ground friction heats the air [B] Air cools at SALR on the wet ascent (releasing latent heat), but warms at the STEEPER DALR during the entire dry descent [C] Solar radiation doubles [D] Coriolis force

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Latent heat released during cloud formation on ascent warms the parcel; during descent, cloud dissipation allows dry adiabatic warming ($3^{\circ}\text{C}/1000\text{ ft}$) over the full height.

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Chapter 6: Global Jet Streams, Clear Air Turbulence & Tropopause

Syllabus Source: CAE Oxford (pp. 175-208) | Pass Target: $\geq 90\%$ AviationExam

1. JET STREAM ARCHITECTURE & CRITERIA

A Jet Stream is a flat, tubular current of quasi-horizontal wind characterized by strong vertical and lateral wind shears:

- ICAO Official Definition: A high-altitude wind stream with a minimum CORE SPEED of at least 60 KNOTS (typically 100 to 200+ knots)!
- Polar Front Jet (PFJ): Located along the Polar Front between polar and tropical air masses. Core height: ~30,000 ft (FL 300 / 300 hPa). Discontinuous, meanders from 40° to 60° latitude. Strongest in winter!
- Subtropical Jet (STJ): Located between Hadley and Ferrel cells at ~30° latitude. Core height: ~40,000 ft (FL 400 / 200 hPa). More stable in position, blowing from West to East.
- Easterly Tropical Jet: Summer jet in the Northern Hemisphere flowing from East to West over southern Asia and central Africa at FL 450 to FL 500.

2. CLEAR AIR TURBULENCE (CAT) DYNAMICS

CAT occurs in cloud-free air, primarily near jet streams and the tropopause break:

- Location of Maximum CAT: Severe CAT is concentrated on the COLD (POLAR) SIDE of the jet core and directly ABOVE the jet core, where horizontal and vertical shears are greatest!
- Critical Shear Criteria: Severe CAT is expected when: Vertical Shear ≥ 6 knots per 1,000 ft, OR Horizontal Shear ≥ 18 knots per 100 NM!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Evaluating Wind Shear for Severe CAT Encounter

SCENARIO (AviationExam High-Altitude Navigation Drill): An airliner is cruising at FL 350 on an airway near the Polar Front Jet: - At FL 350, wind is 270° at 140 knots - The pilot is cleared to descend to FL 310 - At FL 310 (4,000 ft below), wind is reported as 270° at 170 knots (near the jet core) QUESTION: What is the vertical wind shear per 1,000 ft, and should the crew anticipate severe Clear Air Turbulence (CAT)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Total Wind Speed Difference: - Wind at FL 310 = 170 knots - Wind at FL 350 = 140 knots - Difference in speed = 170 kt - 140 kt = 30 knots. Step 2: Calculate the Altitude Difference in Thousands of Feet: - Altitude difference = 35,000 ft - 31,000 ft = 4,000 ft = 4.0 thousand ft. Step 3: Calculate Vertical Wind Shear per 1,000 ft: - Vertical Shear = Speed Difference / Height Difference - Vertical Shear = 30 knots / 4.0 thousand ft = 7.5 knots per 1,000 ft! Step 4: Compare with the ICAO Severe CAT Threshold: - Threshold for Severe CAT = 6.0 knots per 1,000 ft. - Actual Vertical Shear = 7.5 knots per 1,000 ft. - Check: 7.5 kt/1,000 ft > 6.0 kt/1,000 ft -> EXCEEDS SEVERE THRESHOLD! FINAL ANSWER: Vertical shear is 7.5 kt/1,000 ft. Severe CAT is highly probable! The crew must switch on Fasten Seatbelt signs and select Turbulent Air Penetration Speed (V_{ra}).

[!] EXAM TRAP (AVIATIONEXAM) - Cold Side vs Warm Side CAT Trap

- The COLD (low pressure/polar) side of the jet stream has MUCH STRONGER SHEAR than the warm side! - If flying parallel to a westerly jet stream in the Northern Hemisphere: - North of the jet core = Cold side = MAXIMUM TURBULENCE. - South of the jet core = Warm side = Moderate/light turbulence.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the official ICAO minimum core wind speed required for a high-altitude wind current to be classified as a jet stream?

[A] 40 knots [B] AT LEAST 60 KNOTS [C] 100 knots [D] 150 knots

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Under WMO and ICAO definitions, a jet stream must have a minimum wind speed of 60 knots along its central axis.

Q2: In relation to the jet stream core, where is the most severe Clear Air Turbulence (CAT) typically located?

[A] Exactly inside the core where speed is fastest [B] On the COLD (polar) side of the core and just above the core, where wind shear is greatest [C] On the warm side below the core [D] Equatorward of the jet

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Turbulence is generated by wind shear (speed gradient), not absolute speed. The polar side exhibits much tighter horizontal and vertical gradients.

Q3: What seasonal change occurs in the strength and geographic position of the Polar Front Jet (PFJ)?

[A] Strongest in summer and moves equatorward [B] STRONGEST IN WINTER and moves EQUATORWARD (lower latitudes) [C] Strongest in summer over the poles [D] Unchanged throughout the year

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In winter, the equator-to-pole temperature contrast is at its maximum, intensifying the pressure gradient and driving the jet stream further south with higher core speeds.

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 7: Cloud Classification (10 Genera) & Fog Types

Syllabus Source: CAE Oxford (pp. 209-248) | Pass Target: >= 90% AviationExam

1. THE 10 CLOUD GENERA (ICAO CLASSIFICATION)

Clouds are classified by height of base and physical appearance into 10 official genera:

Cloud Level & Height Range Official Cloud Genera Codes		Physical Characteristics & Composition
Cloud Level & Height Range	Official Cloud Genera Codes	Physical Characteristics & Composition
High Clouds (Base > 20,000 ft / FL 200)	Ci (Cirrus), Cc (Cirrocumulus), Cs (Cirrostratus)	Composed purely of ICE CRYSTALS. Cs causes optical HALO around sun/moon. Ci mares' tails signal approaching warm front.
Medium Clouds (Base 6,500 to 20,000 ft)	Ac (Alto cumulus), As (Altostratus), Ns (Nimbostratus)	Mixed water droplets and ice. As gives watery sun appearance. Ns produces continuous rain/snow (dark and thick).
Low Clouds (Base 0 to 6,500 ft AGL)	Sc (Strato cumulus), St (Stratus), Cu (Cumulus)	Water droplets. St is a uniform grey layer causing drizzle. Sc is a rolling grey sheet.
Clouds with Vertical Development	Cb (Cumulonimbus), TCU (Towering Cumulus)	Massive vertical anvil cloud. Generates violent thunderstorms, severe icing, hail, lightning, and microbursts!

2. FOG FORMATION MECHANISMS & DIFFERENCES

- **Radiation Fog:** Formed over land on CLEAR, CALM NIGHTS with light wind (2 to 8 kt) and high RH. Terrestrial radiation chills surface air below dew point. Disperses mid-morning with solar heating or strong wind (> 10 kt mixes and lifts it into Stratus). NEVER OCCURS OVER THE OPEN SEA!
- **Advection Fog:** Formed when WARM, MOIST AIR moves over a COLDER surface. Requires moderate wind (10 to 15 kt). Extremely persistent day and night, over both land and sea (e.g. San Francisco, English Channel).
- **Upslope Fog (Hill Fog):** Formed when moist air is forced up mountain slopes, cooling adiabatically to its dew point.
- **Steam Fog (Arctic Sea Smoke):** Formed when extremely COLD, DRY AIR blows over WARM WATER. Water evaporates rapidly into the cold air, instantly condensing like steam.

3. FOG VS MIST VS HAZE

- **Fog (FG):** Visibility < 1,000 meters, Relative Humidity ~ 100% (water droplets).
- **Mist (BR):** Visibility 1,000 to 5,000 meters, Relative Humidity >= 95% (water droplets).
- **Haze (HZ):** Visibility <= 5,000 meters, Relative Humidity < 95% (dry dust, smoke, pollutants).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Which of the following conditions are REQUIRED for the formation of RADIATION FOG?

[A] Strong wind and thick overcast cloud [B] A CLEAR SKY at night, high relative humidity, and LIGHT WIND (2 to 8 knots) [C] Warm moist air blowing over cold ocean currents [D] Freezing rain

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Radiation fog requires maximum terrestrial cooling (clear sky) and light breeze (2-8 kt) to mix the cooled air throughout the shallow surface layer without dispersing it.

Q2: Why can RADIATION FOG never form over the open ocean?

[A] Sea water contains salt [B] The sea surface temperature undergoes virtually NO DIURNAL TEMPERATURE CHANGE (only $\sim 0.5^{\circ}\text{C}$ to 1°C) [C] Winds over the sea are always calm [D] Too much evaporation

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The high specific heat capacity of water prevents the sea surface from cooling rapidly at night, making radiation fog impossible over the open sea.

Q3: Which cloud genus is responsible for producing an optical HALO around the sun or moon?

[A] Altostratus [B] CIRROSTRATUS (Cs), due to light refraction through hexagonal ice crystals [C] Stratocumulus [D] Nimbostratus

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Cirrostratus is composed entirely of ice crystals that act as prisms, refracting sunlight or moonlight at a 22° angle to produce a prominent halo.

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Chapter 8: Precipitation Physics & Freezing Rain (FZRA)

Syllabus Source: CAE Oxford (pp. 249-278) | Pass Target: $\geq 90\%$ AviationExam

1. MECHANISMS OF PRECIPITATION GROWTH

Cloud droplets are microscopic (~ 0.02 mm) and held aloft by updrafts. Precipitation requires droplet growth by orders of magnitude (~ 1 to 5 mm):

- Collision-Coalescence Process: Dominant in 'warm clouds' (temperatures $> 0^{\circ}\text{C}$, tropics/maritime). Larger droplets fall faster, colliding with smaller droplets and sweeping them up to form raindrops.
- The Bergeron-Findeisen Process: Dominant in 'cold clouds' (temperatures -10°C to -40°C) containing both supercooled water droplets and ice crystals. The saturation vapor pressure over ice is LOWER than over water. Vapor diffuses from water droplets onto ice crystals, causing rapid crystal growth into snowflakes!

2. FORMS OF PRECIPITATION

- Drizzle (DZ): Very small water droplets (diameter < 0.5 mm) falling close together. Produced EXCLUSIVELY by Stratus (St) clouds!
- Rain (RA): Liquid water droplets (diameter 0.5 to 5.0 mm). Continuous rain from Nimbostratus (Ns); Intermittent showers from Cumuliform clouds.
- Sleet / Ice Pellets (PL): Translucent frozen raindrops formed when rain falls through a deep freezing layer near the ground.
- Hail (GR): Balls of concentric ice (diameter ≥ 5 mm). Produced EXCLUSIVELY by Cumulonimbus (CB) with violent updrafts!

3. FREEZING RAIN (FZRA) - THE EXTREME HAZARD

Freezing rain is one of the most hazardous meteorological phenomena in aviation:

- How FZRA Forms: Snow falls from high clouds into a WARM LAYER ALOFT with temperatures $> 0^{\circ}\text{C}$, melting completely into rain. The rain then falls into a SUB-ZERO SURFACE COLD AIR LAYER (temperature $< 0^{\circ}\text{C}$). The drops become supercooled liquid.
- Impact on Aircraft: Upon impacting the cold airframe, the supercooled rain spreads back over the wing and freezes instantly into CLEAR ICE (Glaze). It can accumulate at over 5 cm per hour, covering aerofoils, freezing flight controls, and causing immediate aerodynamic stall!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the Bergeron-Findeisen precipitation process, why do ice crystals grow rapidly while supercooled water droplets do not?

[A] Ice is warmer than water [B] The saturation vapor pressure over ICE is LOWER than the saturation vapor pressure over WATER at the same sub-zero temperature [C] Updrafts only lift ice [D] Sun rays melt water

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Because saturation vapor pressure is lower over ice, a vapor pressure gradient exists from water to ice. Water droplets evaporate and deposit directly onto ice crystals.

Q2: Under which vertical temperature profile does FREEZING RAIN (FZRA) form?

[A] Temperature permanently below -20°C from surface to cruise [B] A WARM LAYER ALOFT with temperature $> 0^{\circ}\text{C}$ situated above a cold freezing surface layer with temperature $< 0^{\circ}\text{C}$ [C] Standard ISA lapse rate [D] Surface heating

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Falling snow melts into liquid rain in the warm air wedge aloft, then becomes supercooled as it falls through the sub-zero layer near the ground, freezing on contact.

Q3: Which cloud type is EXCLUSIVELY responsible for producing DRIZZLE?

[A] Cirrus [B] STRATUS (St) [C] Cumulonimbus [D] Altocumulus

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Drizzle consists of uniform droplets $< 0.5\text{ mm}$ in diameter, produced exclusively by low-level, stable Stratus clouds.

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Chapter 9: Air Masses & Frontal Systems (Warm, Cold, Occluded)

Syllabus Source: CAE Oxford (pp. 279-322) | Pass Target: >= 90% AviationExam

1. AIR MASSES & SOURCE REGIONS

An air mass is a large body of air whose temperature and humidity characteristics are relatively uniform over thousands of kilometers:

- Source Classifications: Arctic (A), Polar (P), Tropical (T); subdivided into Maritime (m) (moist) and Continental (c) (dry). E.g. mP = Maritime Polar (cool, moist, unstable); cT = Continental Tropical (hot, dry, unstable).

2. FRONTAL SYSTEMS COMPARISON

Frontal Parameter	Warm Front (Warm air replaces cold)	Cold Front (Cold air replaces warm)
Frontal Parameter	Warm Front (Warm air replaces cold)	Cold Front (Cold air replaces warm)
Frontal Slope & Speed	Shallow slope (1:150). Slower speed (~15 to 20 kt).	Steep slope (1:50). Fast speed (~25 to 40 kt).
Cloud Sequence on Approach	Ci -> Cs (halo) -> As (watery sun) -> Ns (continuous rain).	Cu building rapidly into Cb and squall lines.
Precipitation Type	Continuous, widespread rain or snow. Extensive low Stratus.	Heavy showers, hail, thunderstorms, brief intense downpour.
Wind Behavior during Passage	Veers (clockwise in N Hemisphere) and decreases slightly.	Sharp, violent VEER (e.g. 210° to 300°), gusty with squalls.
Post-Frontal Weather	Warm sector: High dew point, poor visibility, mist, low drizzle.	Cold air: Rapid clearing, marked temperature DROP, excellent visibility!

3. OCCLUDED FRONTS (COLD VS WARM OCCLUSION)

Because cold fronts move faster than warm fronts, the cold front eventually catches up with the warm front, lifting the warm sector completely off the ground:

- Cold Occlusion: The air behind the advancing cold front is COLDER than the retreating cold air ahead of the warm front. The advancing cold wedge burrows beneath all air masses.
- Warm Occlusion: The air ahead of the warm front is colder than the air behind the cold front. The advancing cold air rides up over the very cold retreating air.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the typical cloud sequence observed ahead of an approaching WARM FRONT?

[A] Cb, TCU, Cu, St [B] CIRRUS (Ci), CIRROSTRATUS (Cs), ALTOSTRATUS (As), and NIMBOSTRATUS (Ns) [C] Stratus, fog, clear sky [D] Altocumulus lenticularis

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The gentle slope (1:150) of a warm front pushes warm air aloft up to 600 NM ahead of the surface front, creating a descending cloud shield: Ci -> Cs -> As -> Ns.

Q2: What happens to the wind direction in the Northern Hemisphere during the passage of an active COLD FRONT?

[A] Backs (turns counter-clockwise) [B] SHARPLY VEERS (turns clockwise, e.g. from Southwest to Northwest) [C] Remains perfectly constant [D] Drops to calm

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Cold front passage is marked by a sharp clockwise wind shift (veering) as the trough line crosses, typically shifting from SW to WNW or NW.

Q3: What characterizes the weather in the COLD AIR MASS immediately behind an active cold front?

[A] Continuous low stratus and poor visibility [B] RAPID CLEARING with isolated convective showers, dropping temperature, and EXCELLENT VISIBILITY [C] Warm sector mist [D] Severe fog

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The cold, dense polar air is dry and clean, resulting in rapid pressure rise, gusty winds, dropping dew points, and superb visibility outside convective showers.

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Chapter 10: Pressure Systems: Depressions, Anticyclones & Col

Syllabus Source: CAE Oxford (pp. 323-358) | Pass Target: >= 90% AviationExam

1. LOW PRESSURE SYSTEMS (DEPRESSIONS / CYCLONES)

A depression is an area of low barometric pressure surrounded by closed isobars. In the Northern Hemisphere, wind circulates COUNTER-CLOCKWISE and inwards towards the center:

- Polar Front Depressions: Form along the boundary between cold polar and warm tropical air. Life cycle: Initial wave -> Developing depression -> Mature occlusion -> Decaying low.
- Thermal Depressions (Heat Lows): Form over hot continental land masses in summer due to intense solar heating (e.g. Iberian Peninsula, Persian Gulf). Stationary, shallow, dissipating at night.
- Orographic Depressions (Lee Lows): Form on the leeward side of mountain ranges as upper air crosses the ridge (e.g. Genoa Low south of the Alps).

2. HIGH PRESSURE SYSTEMS (ANTICYCLONES)

An anticyclone is an area of high barometric pressure with widely spaced isobars. Wind circulates CLOCKWISE and outwards in the Northern Hemisphere:

- Characteristics of Anticyclones: SUBSIDENCE: Air gently sinks from high levels, warming adiabatically. Sinking air suppresses vertical convection, producing stable, cloudless skies at upper levels.
- Subtropical Anticyclones: Permanent warm-core highs (e.g. Azores High) driving global trade winds.
- Cold Continental Anticyclones: Winter highs over snow-covered land (Siberian High, Canadian High). Extremely cold, dense surface air with intense ground inversions.
- Blocking Highs: Large, stagnant high-pressure cells that deflect mid-latitude depressions for weeks, causing persistent weather patterns.

3. RIDGES, TROUGHS & COLS

- Ridge of High Pressure: An elongated wedge of high pressure extending from an anticyclone. Brings improved, settled weather.
- Trough of Low Pressure: An elongated area of low pressure. Acts like a cold front, producing cloud lines, squalls, and showers.
- Col: The neutral saddle-point region between two highs and two lows. Characterized by calm or very light winds. Highly prone to RADIATION FOG IN WINTER, and SEVERE THUNDERSTORMS IN SUMMER!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What weather conditions are typically associated with a 'COL' during summer afternoons over land?

[A] Gale-force winds [B] Light winds and heavy SEVERE THUNDERSTORMS caused by daytime surface heating [C] Persistent radiation fog [D] Snow flurries

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In summer, the calm winds and stagnation of a col allow strong diurnal surface heating to trigger violent convective thunderstorms.

Q2: In an ANTICYCLONE, why is cloud formation generally suppressed at middle and high altitudes?

[A] Strong winds blow clouds away [B] Because of AIR SUBSIDENCE (sinking air warms adiabatically, lowering relative humidity) [C] Air is too cold to form clouds [D] Pressure is too low

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Sinking air warms at DALR (3°C/1,000 ft), moving further away from saturation (lower relative humidity) and evaporating cloud droplets.

Q3: What is a 'BLOCKING HIGH' in European meteorology?

[A] A small thunderstorm cell [B] A large, quasi-stationary anticyclone that remains in place for days or weeks, blocking and deflecting approaching Atlantic depressions [C] An inversion at 1,000 ft [D] A hurricane eye

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Blocking anticyclones divert the mid-latitude westerly storm track to the north or south, locking the region into persistent dry or cold weather.

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Chapter 11: Flight Hazards I: Icing, Turbulence & Mountain Waves

Syllabus Source: CAE Oxford (pp. 359-398) | Pass Target: >= 90% AviationExam

1. AIRFRAME STRUCTURAL ICING PHYSICS

Structural icing occurs when an aircraft flies through visible liquid moisture (clouds/rain) at temperatures below 0°C:

Icing Type	Cloud Type & Temperature Range	Physical Structure & Aerodynamic Severity
Icing Type	Cloud Type & Temperature Range	Physical Structure & Aerodynamic Severity
Rime Ice (Rough / Opaque)	Stratiform clouds (St, Sc, As) at -10°C to -20°C. Small supercooled droplets.	Freezes instantly upon impact without spreading. Traps air, milky white, brittle. Alters aerofoil camber, increases drag.
Clear Ice (Glaze / Transparent)	Cumuliform clouds (Cu, Cb) and FZRA at 0°C to -10°C. Large supercooled droplets.	Freezes slowly, spreading aft beyond de-icing boots. Heavy, hard, tenacious. SEVEREST ICING HAZARD in aviation!
Mixed Ice	Mixed clouds at -10°C to -15°C. Combination of small and large droplets.	Rough, hard, difficult to shed with de-icing boots.
Hoar Frost	Forms on parked aircraft overnight, or descending rapidly into warm, moist air.	Crystalline frost. Degrades boundary layer: INCREASES STALL SPEED by up to 33% and reduces lift by 30%!

2. MOUNTAIN WAVES (LEE WAVES)

Mountain waves develop on the leeward side of mountain ridges under specific conditions:

- Necessary Conditions: 1. Wind direction within 30° of perpendicular to ridge; 2. Wind speed at crest level >= 20 knots and increasing with altitude; 3. Very stable air layer sandwiched between less stable layers above and below.
- Wave Cloud Features: Cap cloud (Föhn wall) over crest; Lenticular clouds (Alto cumulus lenticularis) stationary at wave crests; Rotor clouds (Roll clouds) formed beneath wave crests with VIOLENT, DESTRUCTIVE TURBULENCE!
- Wave Wavelength Rule of Thumb: Wavelength (NM) ~ 0.5 x Wind speed across ridge (knots) (e.g. 40 kt wind -> ~20 NM wavelength).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Mountain Wave Rotor & Descent Hazards

SCENARIO (Mountain Flight Safety Drill): A twin-engine aircraft is flying across the Pyrenees: - Mountain ridge height = 8,000 ft MSL - Wind at crest = 010° at 40 knots (directly perpendicular to ridge) - The leeward side exhibits Altocumulus lenticularis clouds at 12,000 ft and ragged roll clouds at 5,000 ft AGL QUESTION: What is the estimated distance between wave crests, and what lethal hazard lies inside the roll clouds? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Mountain Wave Wavelength: - Formula: Wavelength (NM) $\sim 0.5 \times \text{Wind Speed (knots)}$ - Wavelength = $0.5 \times 40 \text{ kt} = 20 \text{ Nautical Miles!}$ - (Successive downdrafts and updrafts will repeat every 20 NM downwind of the mountain!). Step 2: Identify the Hazards in the Roll Cloud (Rotor Zone): - The roll cloud marks a closed atmospheric vortex (ROTOR). - Air rotates violently about a horizontal axis. - Updrafts and downdrafts exceed 3,000 to 5,000 ft/min. - Flying into a rotor zone can cause structural overload, control loss, and violent barometric altimeter fluctuations! Step 3: Pilot Action: - Never attempt to penetrate roll clouds! Fly at least 3,000 to 5,000 ft ABOVE the mountain crests, and approach ridges at a 45° angle to permit a retreat turn into the valley.

[!] EXAM TRAP (AVIATIONEXAM) - Hoar Frost Lift-Off Trap

- Even a paper-thin layer of hoar frost on the upper wing surface ruins the laminar boundary layer! - It INCREASES STALL SPEED BY UP TO 33%, and REDUCES MAXIMUM LIFT BY 30%! - Take-off with frost on critical surfaces is STRICTLY ILLEGAL (Clean Aircraft Concept).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under what cloud and temperature conditions is SEVERE CLEAR ICE most likely to form on an aircraft?**

[A] Thin Cirrostratus at -40°C [B] CUMULIFORM clouds (Cu, Cb) with large supercooled droplets at temperatures between 0°C and -10°C [C] Fog at +5°C [D] Stratocumulus at -30°C

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Large supercooled droplets have high kinetic energy; upon impact, latent heat of fusion prevents instant freezing, allowing water to flow back and form solid, dense clear ice.

Q2: What atmospheric conditions are ESSENTIAL for the formation of marked MOUNTAIN WAVES?

[A] Calm winds and thick fog [B] Wind blowing nearly PERPENDICULAR to the ridge at ≥ 20 knots, increasing with altitude, and a STABLE atmospheric layer over the crest [C] Intense thunderstorm activity [D] Strong tailwind on landing

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The stable layer acts as an elastic spring: air displaced over the ridge oscillates downstream, creating stationary gravity waves.

Q3: How does a small accumulation of frost on the wing surface affect aerodynamic performance during take-off?

[A] Decreases stall speed [B] Increases drag and DECREASES MAXIMUM LIFT significantly (can cause premature stall on rotation) [C] Has zero effect [D] Increases ground roll friction only

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Frost acts like coarse sandpaper, tripping laminar airflow into turbulent flow early, dramatically reducing maximum lift coefficient $C_{l_{max}}$.

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Chapter 12: Flight Hazards II: Thunderstorms & Microbursts

Syllabus Source: CAE Oxford (pp. 399-438) | Pass Target: >= 90% AviationExam

1. THE THUNDERSTORM (CB) LIFE CYCLE

A thunderstorm requires: (1) High atmospheric instability, (2) Abundant moisture, and (3) A lifting trigger (thermal, frontal, or orographic):

- 1. Cumulus Stage (Initial / Building): Purely UPDRAFTS (1,000 to 3,000 ft/min). Cloud grows vertically into a towering mass. No precipitation reaching ground.
- 2. Mature Stage (Peak Hazard): Marked by the ONSET OF PRECIPITATION at the surface! Updrafts (up to 6,000 ft/min) and downdrafts (up to 4,000 ft/min) coexist side-by-side. Severe turbulence, hail, lightning, microbursts. Duration: 20 to 30 minutes.
- 3. Dissipating Stage (Decay): DOWNDRAFTS PREDOMINATE throughout the cloud. The top freezes into a fibrous cirrus anvil (incus). Rain eases; cloud gradually dissipates.

2. THE MICROBURST PHENOMENON

A microburst is an extremely concentrated, violent downdraft descending from a convective cloud:

- Dimensions & Strength: Horizontal diameter < 4 km (2.2 NM). Downdraft speed can exceed 6,000 ft/min. Lifespan: 5 to 15 minutes.
- Aviation Flight Path Impact: An aircraft flying through a microburst encounters three rapid stages: (1) Severe HEADWIND increase (airspeed jumps, plane balloons above glide slope), (2) Severe DOWNBURST (vertical drop), and (3) Severe TAILWIND increase (airspeed plummets, catastrophic loss of lift and ground impact)!

3. AIRBORNE WEATHER RADAR (AWR) AVOIDANCE RULES

- Lateral Avoidance Margins: Avoid active CB cells by AT LEAST: 20 NAUTICAL MILES at or above FL 200; 10 NM below FL 200 (5 NM if temperature > 0°C).
- Vertical Avoidance: Avoid overflying an active CB by at least 1,000 ft for every 10 knots of wind speed at cloud top (or at least 5,000 ft above anvil)!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What meteorological event formally marks the transition from the Cumulus stage to the MATURE STAGE of a thunderstorm?

[A] The first formation of an anvil [B] The ONSET OF PRECIPITATION reaching the ground and the appearance of downdrafts [C] Lightning seen in the distance [D] Cloud top reaching FL 100

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Friction from falling rain drops drags air down, creating the first downdraft and marking the start of the mature stage.

Q2: When penetrating an active MICROBURST on final approach, what is the initial indication observed in the cockpit?

[A] Sudden severe loss of airspeed [B] A SUDDEN INCREASE IN HEADWIND causing an abrupt increase in indicated airspeed and pitch-up above the glide path [C] Stall warning [D] Altimeter drops to zero

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The outflow ring produces a headwind first, causing airspeed to jump. If the pilot reduces thrust, the subsequent downdraft and tailwind will cause an unrecoverable crash.

Q3: What is the recommended minimum lateral separation from an active Cumulonimbus (CB) cell at FL 250?

[A] 5 NM [B] AT LEAST 20 NAUTICAL MILES [C] 2 NM [D] 1 NM

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. At high flight levels (above FL 200), hail can be ejected miles outside the visible cloud column; 20 NM lateral clearance is standard airline SOP.

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Chapter 13: Tropical Meteorology, Revolving Storms & Climatology

Syllabus Source: CAE Oxford (pp. 439-472) | Pass Target: $\geq 90\%$ AviationExam

1. THE INTERTROPICAL CONVERGENCE ZONE (ITCZ)

The ITCZ is the equatorial meteorological trough where the Northeast Trade Winds of the Northern Hemisphere meet the Southeast Trade Winds of the Southern Hemisphere:

- Characteristics: Intense solar heating causes broad-scale convergence and ascent, generating massive clusters of Cumulonimbus (CB) reaching FL 550 to FL 600. Known maritime name: The Doldrums.
- Seasonal Migration: Migrates north of the equator in Northern summer (July) and south in Northern winter (January). Migration is much greater over land masses (up to 25°N over Asia) than over oceans.

2. TROPICAL REVOLVING STORMS (TRS)

Severe tropical cyclones with warm cores and sustained winds ≥ 64 knots. Names: Hurricane (Atlantic/Eastern Pacific), Typhoon (Northwest Pacific), Cyclone (Indian Ocean/South Pacific):

- Mandatory Formation Conditions: 1. Warm ocean water with Sea Surface Temperature (SST) $\geq 26^{\circ}\text{C}$ to 27°C down to 50 m depth; 2. Latitude between 5° and 20° (Coriolis force is too weak to create rotation between 0° and 5° !); 3. Low vertical wind shear (< 10 kt); 4. Pre-existing tropical disturbance.
- TRS Anatomy: The Eye: 10 to 30 NM diameter central core with calm/light winds, descending air (subsidence), clear skies; The Eyewall: Ring of ferocious towering CBs surrounding the eye with MAXIMUM SUSTAINED WINDS (100 to 180 kt) and torrential rain; Storm Surge: Coastal sea level rise responsible for 90% of casualties.

3. THE ASIAN MONSOONS

- Southwest Monsoon (Summer: June to Sept): Intense heating over Asia creates a deep continental heat low. Warm, moisture-laden maritime air from the Indian Ocean sweeps northeast across India, bringing torrential rainfall and continuous low cloud.
- Northeast Monsoon (Winter: Dec to March): The massive Siberian High builds over cold Asia. Cool, dry continental air flows southwest towards the ocean, producing clear skies and dry weather.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why do Tropical Revolving Storms (Hurricanes / Typhoons) NEVER form between latitudes 0° and 5° North?

[A] Ocean water is too cold [B] The CORIOLIS FORCE is too weak near the equator to initiate cyclonic rotation [C] Trade winds are too strong [D] Too many islands

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Coriolis parameter is proportional to $\sin(\text{latitude})$. At latitudes under 5° , Coriolis force is virtually zero, preventing air from spinning into a closed vortex.

Q2: What weather conditions exist inside the 'EYE' of a mature Tropical Revolving Storm?

[A] Maximum hurricane-force winds [B] CALM OR LIGHT WINDS, descending air (subsidence), and relatively CLEAR SKIES surrounded by the towering eyewall [C] Continuous hail [D] Severe turbulence

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Sinking air in the central eye suppresses cloud formation, creating an eerie pocket of calm air surrounded by violent eyewall winds.

Q3: What causes the wet SOUTHWEST MONSOON over the Indian subcontinent between June and September

[A] Cold fronts from Antarctica [B] Intense summer solar heating forming a continental thermal low over Asia, drawing warm, moist maritime air from the Indian Ocean [C] High pressure over the Himalayas [D] Volcanic activity

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Asian thermal low draws equatorial maritime air across the Arabian Sea and Bay of Bengal, unleashing the summer monsoon.

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 14: Meteorological Reports & Forecasts (METAR, TAF, SIGMET)

Syllabus Source: CAE Oxford (pp. 473-510) | Pass Target: >= 90% AviationExam

1. METAR & SPECI DECODING RULES

METAR is an aerodrome routine meteorological report issued every 30 or 60 minutes. SPECI is a special report issued when weather deteriorates across critical thresholds:

- Wind Group: e.g. 24015G28KT = Mean wind 240° True at 15 kt, gusting to 28 kt. 180V260 = Wind direction varying between 180° and 260°.
- Visibility & RVR: e.g. 1400 = Prevailing visibility 1,400 meters. R27/0800VP1500U = Runway 27 RVR varying from 800 m to greater than 1,500 m, with upward trend (U).
- Present Weather Codes: Descriptors: TS (Thunderstorm), FZ (Freezing), SH (Shower); Precipitation: DZ (Drizzle), RA (Rain), SN (Snow), GR (Hail >= 5 mm); Obscuration: FG (Fog), BR (Mist), HZ (Haze).
- Cloud Coverage & Base: FEW = 1-2 octas, SCT = 3-4 octas, BKN = 5-7 octas (CEILING!), OVC = 8 octas (CEILING!). Base given in hundreds of feet AGL (e.g. BKN015 = Broken at 1,500 ft AGL). Only CB and TCU cloud types are appended.
- CAVOK Conditions (All 4 must be satisfied!): 1. Visibility >= 10 km; 2. No cloud below 5,000 ft or Minimum Sector Altitude (MSA); 3. No CB or TCU; 4. No significant weather phenomena!

2. TAF (TERMINAL AERODROME FORECAST)

- Validity Periods: Short TAF: 9 hours (issued every 3h). Long TAF: 24 to 30 hours (issued every 6h).
- Change Groups: BECMG: Gradual change over the specified time window. TEMPO: Temporary fluctuations lasting < 1 hour and covering < 50% of the period. FM (From): Rapid permanent change at the exact hour and minute.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Complete METAR & TAF Flight Dispatch Decoding

SCENARIO (EASA Flight Deck Dispatch Drill): A flight crew reviews the destination weather: METAR LEMD 121430Z 04018G32KT 010V080 3000 +SHRA SCT008 BKN018CB 14/11 Q1008 NOSIG= QUESTION: Decode all items in this METAR step-by-step for flight dispatch. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Header -> LEMD (Madrid Barajas), 12th day of the month at 14:30 UTC. Step 2: Wind -> 04018G32KT = Mean wind 040° True at 18 knots, gusting to 32 knots! - 010V080 = Wind direction is varying clockwise/counter-clockwise between 010° and 080°. Step 3: Visibility -> 3000 = Minimum prevailing visibility is 3,000 meters. Step 4: Weather -> +SHRA = HEAVY (+) SHOWERS (SH) OF RAIN (RA)! Step 5: Clouds: - SCT008 = Scattered (3 to 4 octas) at 800 ft AGL (NOT a ceiling). - BKN018CB = Broken (5 to 7 octas) at 1,800 ft AGL containing CUMULONIMBUS (CB)! This constitutes an official CEILING at 1,800 ft with severe convective hazards! Step 6: Temperatures -> 14/11 = Outside Air Temperature +14°C, Dew Point +11°C (narrow 3°C spread). Step 7: Pressure -> Q1008 = QNH is 1008 hPa. Step 8: Trend -> NOSIG = No significant change expected during the next 2 hours. FINAL ANSWER: Madrid has gusty winds to 32 kt, heavy rain showers, visibility 3 km, and CB cloud ceiling at 1,800 ft AGL.

[!] EXAM TRAP (AVIATIONEXAM) - The Ceiling Definition Trap

- By official ICAO definition, a 'CEILING' is the height above ground of the lowest layer of clouds covering MORE THAN HALF THE SKY: - BKN (5-7 octas) = CEILING. - OVC (8 octas) = CEILING. - FEW (1-2 octas) and SCT (3-4 octas) are NOT ceilings!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Which of the following conditions MUST be met simultaneously for 'CAVOK' to be reported in a METAR?**

[A] Visibility 5,000 m and no low clouds [B] Visibility ≥ 10 km, NO CLOUD below 5,000 ft or MSA (whichever is higher), NO CB or TCU, and NO significant weather [C] Clear sky and zero wind [D] Zero fog

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. CAVOK (Ceiling And Visibility OK) requires ≥ 10 km visibility, absence of clouds below 5,000 ft / MSA, no CB/TCU at any level, and no significant weather phenomena.

Q2: In aviation meteorology, which cloud coverage codes constitute an official 'CEILING'?

[A] Only OVC (Overcast) [B] BROKEN (BKN: 5 to 7 octas) and OVERCAST (OVC: 8 octas) [C] SCT and BKN [D] FEW and SCT

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 3, a cloud ceiling is defined as the height of the base of the lowest layer covering more than 4 octas (i.e. BKN or OVC).

Q3: In a TAF forecast, what does the change indicator 'TEMPO 1216 2000' signify?

[A] Permanent change to 2,000 m visibility from 12:00 to 16:00 [B] TEMPORARY fluctuations to 2,000 meters visibility occurring between 12:00 UTC and 16:00 UTC, each lasting less than 1 hour and covering in aggregate less than half the period [C] Temperature will reach 20°C [D] Wind 200° at 16 kt

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. TEMPO denotes temporary weather fluctuations lasting less than 60 minutes per instance and less than 50% of the total time window.